

Anxiety and associated factors among Vietnamese students during COVID-19 pandemic

A cross-sectional study

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Abstract

Mental health care for students in general, particularly anxiety, is a significant problem that needs more attention, especially during the coronavirus disease 2019 (COVID-19) pandemic. This study aimed to describe the prevalence of anxiety and examine the associated factors among students during the COVID-19 pandemic in Vietnam. A cross-sectional study was conducted from August to September 2021 among 5730 students. An online survey was used to collect sociodemographic information, and the generalized anxiety disorder questionnaire (GAD-7) was used to assess anxiety symptoms among Vietnamese students. Results showed that the prevalence of anxiety among study participants was 16.2% (95% confidence interval [CI]: 15.3%–17.2%). Factors related to anxiety among students were gender, type of housemate, COVID-19 exposure/infection status, vaccination status, health status, academic performance, and social relationships during the COVID-19 pandemic. A significant number of students experienced anxiety during COVID-19, and this rate was related to several factors. Psychological interventions are required to support students during and after the COVID-19 pandemic and other health crises. Further studies are required to confirm our findings.

Abbreviations: CI = confidence interval, COVID-19 = coronavirus disease 2019, GAD-7 = generalized anxiety disorder questionnaire, PR = prevalence ratios.

Keywords: anxiety, COVID-19 pandemic, students, Vietnam

1. Introduction

Before the COVID-19 pandemic hit the world, students' mental health issues had already been identified as the top obstacle to their academic performance. Mental disorders can affect student motivation, concentration, and social interactions. According to a report by Collegiate Mental Health (2019), which included 163 college and university counseling centers from 54 countries in the world, anxiety is the most common mental health problem among students, with 82,685 students (62.7% of the sample) reporting having anxiety symptoms.^[1] Two years after the COVID-19 pandemic first appeared in October 2019, the World Health Organization—WHO called COVID-19 “a pandemic that causes widespread concern and

increases anxiety and stress worldwide.”^[2] The COVID-19 epidemic has caused a wide range of psychological problems, most notably anxiety, panic disorders, depression, obsessive-compulsive disorders, and psychiatric disorders.^[3,4] A 36-study systematic review from September 2020 to February 2021 revealed the prevalence of anxiety to be about 41% (95% confidence interval [CI] = 0.34–0.49).^[5]

University students who have to adjust to different studying conditions during the pandemic also experience psychological problems, most notably, an increase in anxiety and stress. Jiang et al (2021) surveyed 1195 students from 4 countries in the Asia-Pacific region (Malaysia, Indonesia, Thailand, and China). Their study revealed that anxiety was the most common

The authors have no funding and conflicts of interest to disclose.

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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How to cite this article: Nguyen LX, Dao LTD, Ta AN, Le HT, Nguyen HV, Nguyen LTM, Nguyen PT, Nguyen TT, Ta TC, Nguyen TH, Nguyen TT, Huynh TA, Hoang AQ, Duong LTH, Do LH, Pham NT. Anxiety and associated factors among Vietnamese students during COVID-19 pandemic: A cross-sectional study. *Medicine* 2023;102:16(e33559).

Received: 4 January 2023 / Received in final form: 19 March 2023 / Accepted: 28 March 2023

<http://dx.doi.org/10.1097/MD.00000000000033559>

psychological problem among students, followed by depression and stress. Severe and extremely severe anxiety were observed in 20.5% of the participants. An increase in anxiety was also observed in studies conducted in other countries, such as the United States, Sweden, and Turkey. An online survey of 2031 participants revealed that 38.48% of university students in the US had moderate-to-severe levels of anxiety, while 71.26% had increased levels of stress and anxiety during the pandemic.^[6] Students are at risk of mental health problems during the COVID-19 pandemic due to unexpected life changes, academic challenges, and uncertain future employment opportunities.^[7] In the same research direction, Md. Najmol et al (2021) surveyed students in Bangladesh who were studying online because of lockdowns. The results showed that 14.8% experienced extreme levels of anxiety, and 82.5% experienced mild to extreme levels of anxiety.^[8] The study also pointed out several factors that affect anxiety in university students during the COVID-19 pandemic, including gender, family size, residential area, academic year, accessibility to high-speed Internet and online learning platforms, learning alone at home, and lack of access to online learning resources. Among these factors, students' most significant problem when studying online during the COVID-19 pandemic was the limitation of online learning platforms.^[8]

In Vietnam, a few studies have been conducted to study mental health issues during the COVID-19 pandemic in the general population and among healthcare workers and university students.

Understanding the current situation of student anxiety and its related factors under the impact of the COVID-19 pandemic is an urgent and necessary issue. However, there are still very few studies on student anxiety, especially regarding the correlation between student anxiety and several influential factors. This study aimed to describe the prevalence of anxiety and examine the associated factors among students during the COVID-19 pandemic in Vietnam. Our findings could add to the literature regarding anxiety among university students and provide evidence to advocate that stakeholders pay attention to mental health care for students during the COVID-19 pandemic. Moreover, based on these results, researchers can suggest activities to support psychological interventions for students.

2. Methods

2.1. Study design and participants

We conducted an online cross-sectional study among students in Vietnam.

Students were recruited based on the following criteria: being at least 18 years old; living in Vietnam during the survey period; agreeing to participate in the survey by signing an online informed consent document; and accessing the online questionnaire via the Internet.

2.2. Study sample size and sampling

The survey was conducted during a 2-week social distancing and isolation period from August 23 to September 5, 2021. A self-selection sampling technique was used to recruit participants. The link to the online questionnaire was sent to the network of university lecturers in Vietnam, who were required to forward the survey link to their students via Email, Facebook, or Zalo mobile applications. Students were also encouraged to share links widely with other students. In total, 5730 students participated in this study.

2.3. Study variables

2.3.1. Dependent variable. The dependent variable was the anxiety status, measured based on the Generalized Anxiety

Disorder Questionnaire (GAD-7) questionnaire, which assessed the anxiety level experienced by the participants in the past 2 weeks (See the study instrument).

2.3.2. Independent variables. Independent variables included gender, type of housemate, COVID-19 exposure/infection status, COVID-19 vaccination status, current health status, academic performance, and social relationships.

2.4. Study instruments and measurements

There were 2 main parts in the study instruments. The first part included questions about the sociodemographic factors of participants. The second part was GAD-7. The questionnaire assesses the bothering of each symptom over the past 2 weeks. The GAD-7 is an internationally recognized screening tool used for various settings and populations.^[9] Several studies in Vietnam used the Vietnamese version of the GAD-7 to screen for anxiety symptoms among the Vietnamese population, health workers, students, etc., during the COVID-19 pandemic.^[10–12] The GAD-7 includes 7 questions using a 4-point Likert-type rating scale from 0 (not at all) to 3 (nearly every day) to measure the anxiety level experienced by the participants in the past 2 weeks. The total score for all 7 items was used to measure the level of anxiety, with higher scores indicating greater anxiety. The total GAD-7 score was grouped into 4 categories: none (0–4 points), mild (5–9 points), moderate (10–14 points), and severe (15–21 points). At a cutoff point of 10, both sensitivity and specificity exceed 80%.^[13] Previous studies have also used this cutoff point.^[14,15] In our study, participants with scores of 10 and above were identified as screening positive for anxiety. The internal consistency coefficient (Cronbach alpha) on this sample is 0.92.

2.5. Statistical analysis

We used the chi-square test to compare differences in student characteristics. Multiple Poisson regression models with robust error variances were used to examine the factors associated with student anxiety during the COVID-19 pandemic.^[16–18] We used Robust (or modified) Poisson regression models to calculate the Prevalence Ratios (PRs) and corresponding 95% CI (using ROBUST option) and used a significance level of $P < .05$.^[19] All analyses were conducted using Stata 14.2.^[20]

2.6. Ethical consideration

The Institutional Review Board of the Vietnam Association of Psychology approved this study under Decision No. 10/2021/HTLHVN-DD. Informed consent was obtained from all participants before data collection. The participants volunteered, and anonymity was ensured. Respondents could stop or withdraw from the study at any time.

3. Results

3.1. General characteristics of the study participants

Table 1 shows the participants' sociodemographic characteristics. Of the 5730 study participants, 63.9% were female. The majority of them were living with their families (81.8%), had no contact with COVID-19 infected persons (88.5%), received no dose of COVID-19 vaccination (75.5%), and had a stable health status (84.4%). Half of the students reported stable academic performance (50.0%) and social relationships (55.4%).

Table 1
Characteristics of participants.

Factor	Female	Male	Total	
	3667	2063	5730	
N	n (%)	n (%)	n (%)	P value
Types of housemate				.005
Living with family	3032 (82.7)	1655 (80.2)	4687 (81.8)	
Living with relatives	73 (2.0)	45 (2.2)	118 (2.1)	
Living in a dormitory	121 (3.3)	61 (3.0)	182 (3.2)	
Living with friends in a boarding house	278 (7.6)	163 (7.9)	441 (7.7)	
Living alone in a boarding house	163 (4.4)	139 (6.7)	302 (5.3)	
COVID-19 exposure/infection status				.86
Never had contact with COVID-19 infected person	3255 (88.8)	1818 (88.1)	5073 (88.5)	
Had COVID-19 infection	16 (0.4)	8 (0.4)	24 (0.4)	
Had contact with COVID-19 infected person (F0)	37 (1.0)	18 (0.9)	55 (1.0)	
Had contact with COVID-19 infected person (F1)	112 (3.1)	68 (3.3)	180 (3.1)	
Had contact with COVID-19 infected person (F2)	247 (6.7)	151 (7.3)	398 (6.9)	
COVID-19 vaccination status				.22
Never received a dose of COVID-19 vaccination	2756 (75.2)	1572 (76.2)	4328 (75.5)	
Got the first dose of COVID-19 vaccination	724 (19.7)	373 (18.1)	1097 (19.1)	
Got the second dose of COVID-19 vaccination	187 (5.1)	118 (5.7)	305 (5.3)	
Current health status				<.001
Better	206 (5.6)	240 (11.6)	446 (7.8)	
Stable	3215 (87.7)	1642 (79.6)	4857 (84.8)	
Worse	246 (6.7)	181 (8.8)	427 (7.5)	
Academic performance				.48
Better	280 (7.6)	170 (8.2)	450 (7.9)	
Stable	1854 (50.6)	1012 (49.1)	2866 (50.0)	
Worse	1533 (41.8)	881 (42.7)	2414 (42.1)	
Social relationships				<.001
Better	226 (6.2)	161 (7.8)	387 (6.8)	
Stable	2132 (58.1)	1042 (50.5)	3174 (55.4)	
Worse	1309 (35.7)	860 (41.7)	2169 (37.9)	
Self-reported anxiety				<.001
No	3009 (82.1)	1792 (86.9)	4801 (83.8)	
Yes	658 (17.9)	271 (13.1)	929 (16.2)	

COVID-19 = coronavirus disease 2019.

3.2. The prevalence of students having anxiety symptoms

Table 2 shows the prevalence of anxiety symptoms among study participants. 16.2 of the study students, 16.2% reported having anxiety symptoms during the COVID-19 pandemic (95% CI: 15.3%–17.2%). The prevalence of anxiety was statistically significantly higher among female students than male students (17.9% vs 13.1%, $P < .001$). The prevalence of anxiety was also significantly higher among students who lived in a dormitory, lived in a boarding house, and lived alone in a boarding house, as compared to the other groups of the same category.

The prevalence of anxiety was found to be statistically significantly higher among the students who had COVID-19 infection and had contact with COVID-19 infected people (F0, F1, F2).

The prevalence of anxiety was significantly higher among students with poor health status than among those with good health status. The prevalence of anxiety was significantly higher among students with poor academic performance and social relationships than among those with stable academic performance and social relationships.

3.3. Factors associated with self-reported anxiety among students

Table 3 presents the results of the Robust Poisson regression analysis of the factors associated with anxiety symptoms among students. Statistically significant factors related to the anxiety among students included: Gender: The prevalence of anxiety among the male students was lower than that among the female students (PR = 0.66, 95% CI: 0.58–0.74); Types of housemate:

Students were living in a dormitory or those were living alone in a boarding house had higher prevalence of anxiety as compared to students who were living with family (PR = 1.71, 95% CI: 1.39–2.12; PR = 1.36, 95% CI: 1.10–1.69; respectively); COVID-19 exposure/infection status: The prevalence of anxiety among students who had COVID-19 infection or had contact with COVID-19 infected person (F0) was higher than that among the students who had no COVID-19 exposure/infection (PR = 2.29, 95% CI: 1.40–3.75; PR = 2.04, 95% CI: 1.43–2.91; respectively); COVID-19 vaccination status: The prevalence of anxiety among students who had first and second dose administered had lower prevalence of anxiety as compared to students who never received dose (PR = 0.97, 95% CI: 0.95–0.98; PR = 0.95, 95% CI: 0.92–0.98; respectively); The current health status, academic performance, and social relationships: The student with poor health status, poor academic performance and poor social relationships had higher prevalence of anxiety as compared to the other groups of the same category (PR = 1.71, 95% CI: 1.34–2.18; PR = 1.44, 95% CI: 1.12–1.87; PR = 1.54, 95% CI: 1.16–2.04; respectively).

4. Discussion

At the time of this study, Vietnam was in the middle of the fourth virus outbreak, with an average of 12,242 new locally acquired cases per day (435,265 cases in total; 10,749 deaths), and 62/63 cities and provinces had reported COVID-19 cases.^[21] This study examined Vietnamese university students' anxiety levels during the COVID-19 pandemic. Anxiety symptoms were observed in 16.2% of the students. This prevalence is still lower than that

Table 2
Anxiety of participants.

	Self-reported anxiety		P value
	No	Yes	
	4801 (83.8)	929 (16.2)	
	n (%)	N (%)	
Gender			<.001
Female	3009 (82.1)	658 (17.9)	
Male	1792 (86.9)	271 (13.1)	
Types of housemate			<.001
Living with family	3990 (85.1)	697 (14.9)	
Living with relatives	95 (80.5)	23 (19.5)	
Living in a dormitory	123 (67.6)	59 (32.4)	
Living with friends in a boarding house	366 (83)	75 (17)	
Living alone in a boarding house	227 (75.2)	75 (24.8)	
COVID-19 exposure/infection status			<.001
Never had contact with COVID-19 infected person	4290 (84.6)	783 (15.4)	
Had COVID-19 infection	15 (62.5)	9 (37.5)	
Had contact with COVID-19 infected person (F0)	34 (61.8)	21 (38.2)	
Had contact with COVID-19 infected person (F1)	142 (78.9)	38 (21.1)	
Had contact with COVID-19 infected person (F2)	320 (80.4)	78 (19.6)	
COVID-19 vaccination status			.99
Never received dose	3626 (83.8)	702 (16.2)	
First dose administered	920 (83.9)	177 (16.1)	
Second dose administered	255 (83.6)	50 (16.4)	
Current health status			<.001
Better	376 (84.3)	70 (15.7)	
Stable	4167 (85.8)	690 (14.2)	
Worse	258 (60.4)	169 (39.6)	
Academic performance			<.001
Better	393 (87.3)	57 (12.7)	
Stable	2552 (89)	314 (11)	
Worse	1856 (76.9)	558 (23.1)	
Social relationships			<.001
Better	341 (88.1)	46 (11.9)	
Stable	2791 (87.9)	383 (12.1)	
Worse	1669 (76.9)	500 (23.1)	

COVID-19 = coronavirus disease 2019.

in cohorts from other countries, such as India,^[22] Guangdong, China,^[23] Turkey,^[24] and Saudi Arabia.^[25] These findings were quite similar to the results of another study that assessed anxiety levels in the Vietnamese population, which showed that Vietnamese people had lower anxiety levels (compared to people in China and Iran) because of Vietnam success in reducing the spread of COVID-19 in the country due to the direction of the Vietnamese Government and proactive and effective community health actions.^[26] Thus, in addition to depression - a common mental health disorder of students during the pandemic,^[27-29] anxiety is also a common psychological problem of students during this period. Anxiety can lead to changes in sleep patterns, and those with higher levels of anxiety are related to a lower grade point average and experience of COVID-19 symptoms. Moreover, students with lower grade point average and prior experience with COVID-19 symptoms were more likely to feel depressed.^[27]

In our regression models, we found that gender was a factor associated with self-reported anxiety, with women having higher anxiety levels than men. This result was consistent with previous findings,^[27,30,31] which found higher levels of anxiety in female students than in male students. Therefore, it can be suggested that gender may be a predictor of anxiety because of personality traits. Females are higher levels of neuroticism or emotional instability/negative emotionality than males.^[32]

In this study, we found that having a fair/poor health status, spending 8 or more hours a day in front of a screen, and having a contact history with a positive COVID-19 case predicted more psychological issues when risk factors were considered simultaneously.^[33] In this study, self-reported health status was a predictor of higher anxiety

levels among students. Students who confirmed their health was worse than before were more likely to worry than others.^[34]

Similar to previous studies, contact history with COVID-19 positive cases was also a significant predictor of higher levels of anxiety in university students.^[35,36] At the time of this study, the number of infections per day in Vietnam was extremely high at 87,206 new cases and 2472 deaths in 1 week.^[21] In addition, the rapid spread of information about the pandemic on social networks contributes to the increase in students' anxiety levels.^[37] Thus, fear of infection has been cited as a stressor that leads to anxiety and depression.^[38]

Our research showed that vaccinated students had a lower prevalence of anxiety than unvaccinated students. There is evidence that vaccinated students place greater value on their health and thus are more inclined to be vaccinated. Therefore, it can be assumed that vaccines have a protective role in mental and physical health.^[39]

The results also showed that participants who rated their academic performance and quality of social relationships as "worsened" had approximately 1.2 times higher levels of anxiety compared to those who rated these domains as "unchanging" or "got better." Previous research has also shown that the self-rated quality of social relationships influences university students' mental health problems, including anxiety.^[24,40] For example, Cam et al (2021) identified female sex and poor family relationships as risk factors for PTSD, anxiety, depression, and stress. Anxiety is significantly associated with family factors, such as family conflict, obligations, and expectations.^[40] This finding indicates that the quality of students' relationships, including family relationships, may be the reason for their higher anxiety levels. For the relationship between academic results and

Table 3
Factors associated with anxiety.

	PR	95% CI	
Gender			
Female	REF		
Male	0.87	[0.86–0.88]	**
Types of housemate			
Living with family	REF		
Living with relatives	1.01	[0.97–1.07]	
Living in a dormitory	1.11	[1.06–1.15]	**
Living with friends in a boarding house	1.03	[1.01–1.06]	*
Living alone in a boarding house	1.12	[1.08–1.15]	**
COVID-19 exposure/infection status			
Never had contact with COVID-19 infected person	REF		
Had COVID-19 infection	1.18	[1.07–1.30]	**
Had contact with COVID-19 infected person (F0)	1.23	[1.15–1.31]	**
Had contact with COVID-19 infected person (F1)	1.09	[1.04–1.13]	**
Had contact with COVID-19 infected person (F2)	1.07	[1.04–1.09]	**
COVID-19 vaccination status			
Never received dose	REF		
First dose administered	0.97	[0.95–0.98]	*
Second dose administered	0.95	[0.92–0.98]	*
Current health status			
Better	REF		
Stable	0.95	[0.93–0.98]	*
Worse	1.25	[1.21–1.30]	**
Academic performance			
Better	REF		
Stable	0.97	[0.95–1.00]	
Worse	1.12	[1.09–1.16]	**
Social relationships			
Better	REF		
Stable	0.94	[0.92–0.97]	**
Worse	1.13	[1.09–1.16]	**
N	5730.00		

The ROBUST option in STATA Poisson procedure.

CI = confidence interval, COVID-19 = coronavirus disease 2019, PR = prevalence ratios.

* $P < .01$.

** $P < .001$.

students' anxiety, the study showed that general mental health problems, including anxiety disorders, can affect students' energy levels, concentration, dependability, mental abilities, and optimism, prevent the performance of activities.^[41]

The current study has some limitations. First, it is a cross-sectional study; therefore, the data are only valid for a certain period. Second, our study was conducted using a convenience sample technique, so the results may not be representative enough to apply to all university students across Vietnam. Finally, this study did not include coping or prevention measures. This should be explored further in future studies.

In conclusion, the prevalence of anxiety among students during the COVID-19 pandemic in Vietnam is relatively high. We also found a strong association between anxiety and various factors, such as sex, types of housemates, COVID-19 exposure/infection status, vaccination status, current health status, academic performance, and social relationships during the COVID-19 pandemic. Given these findings, psychological interventions should be applied to students during and after the COVID-19 pandemic and other health crises. Further studies are required to confirm our findings.

Psychological interventions are the methods in which counselors use activities or groups of activities to change behaviors, feelings, and emotional states.^[42] Psychological interventions are variable, with many different shapes and sizes.^[43]

Acknowledgments

The authors would like to thank all colleagues and students who participated in this study for their support.

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